Week 4 – ReactJS

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Exercise 1 (Line 31): Creating Your First React Application

Objective

Create your first React application using **Vite** (recommended) or **Create React App**, and display a simple welcome message.

Step 1: Create the Project

Open Terminal or Command Prompt:

npm create vite@latest my-first-react-app

Choose:

Framework : React
Variant : JavaScript

Move into the project:

cd my-first-react-app

Install dependencies:

npm install

Start the development server:

npm run dev

You will see an address like:

http://localhost:5173

Open it in your browser.

Project Structure

```
my-first-react-app
|
├── node_modules
├── public
├── src
|   ├── App.jsx
|   ├── main.jsx
|   ├── App.css
```

```
|   └── index.css
|
├── package.json
└── vite.config.js
```

Step 2: Edit App.jsx

Replace the default code with:

```
function App() {
  return (
    <div>
      <h1>Welcome to React!</h1>
      <h2>My First React Application</h2>
      <p>Hello, Java FSE Learner!</p>
    </div>
  );
}

export default App;
```

Step 3: main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

Output

When you open:

http://localhost:5173

You should see:

Welcome to React!

My First React Application

Hello, Java FSE Learner!

How React Works

```
Browser
  |
  ▼
main.jsx
  |
  ▼
<App />
  |
  ▼
App.jsx
  |
  ▼
HTML rendered on screen
```

Important Concepts

What is React?

React is a **JavaScript library** used to build interactive user interfaces using reusable components.

What is JSX?

JSX (**JavaScript XML**) allows you to write HTML-like syntax inside JavaScript.

Example:

```
const element = <h1>Hello React!</h1>;
```

What is a Component?

A component is a reusable piece of UI.

Example:

```
function App() {
  return <h1>Hello</h1>;
}
```

Exercise 2 (Line 32): React Components and Props

This is one of the most important React concepts and is frequently asked in interviews.

Objective

Learn to:

- Create reusable React Components

- Pass data using **Props**
- Render multiple components

Project Structure

```
ReactComponentsDemo
|
├── src
|   ├── components
|   |     Student.jsx
|   |
|   ├── App.jsx
|   ├── main.jsx
|   └── index.css
|
├── package.json
└── vite.config.js
```

Step 1: Create Student Component

Create a folder:

src/components

Create **Student.jsx**

```
function Student(props) {

  return (
    <div style={{
      border: "2px solid black",
      padding: "15px",
      margin: "10px"
    }}>

      <h2>Student Details</h2>

      <p><b>Name :</b> {props.name}</p>

      <p><b>Age :</b> {props.age}</p>

      <p><b>Course :</b> {props.course}</p>

    </div>
  );

}
```

```
export default Student;
```

Step 2: App.jsx

```
import Student from "./components/Student";

function App() {

  return (

    <div>

      <h1>React Components and Props Demo</h1>

      <Student
        name="Harshpreet Singh"
        age="23"
        course="Java FSE"
      />

      <Student
        name="Rahul"
        age="22"
        course="ReactJS"
      />

      <Student
        name="Aman"
        age="24"
        course="Spring Boot"
      />

    </div>

  );

}

export default App;
```

Step 3: main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
```

```
import App from "./App";

ReactDOM.createRoot(document.getElementById("root")).render(

  <React.StrictMode>
    <App />
  </React.StrictMode>

);
```

Output

```
React Components and Props Demo

-------------------------------
Student Details

Name : Harshpreet Singh
Age : 23
Course : Java FSE

-------------------------------

Student Details

Name : Rahul
Age : 22
Course : ReactJS

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Student Details

Name : Aman
Age : 24
Course : Spring Boot
```

Each **Student** component displays different data while using the same component code.

How Props Work

```
App.jsx
  |
  |
  ├── name="Harshpreet"
  ├── age="23"
  └── course="Java FSE"
```

```
        |
        ▼
Student Component
        |
Reads
props.name
props.age
props.course
        |
        ▼
Displays Data
```

What are Props?

Props (Properties) are used to pass data from a **Parent Component** to a **Child Component**.

Example:

```
<Student
   name="Rahul"
   age="22"
   course="ReactJS"
/>
```

Inside the child component:

```
props.name
props.age
props.course
```

Parent and Child Components

Parent	Child
App.jsx	Student.jsx

The parent passes data, and the child receives it through props.

Why Use Components?

- Reusable code
- Better organization
- Easier maintenance
- Faster development

Instead of writing the student details three times, we create one component and reuse it with different props.

Alternative: Destructuring Props

Instead of:

```
function Student(props) {

  return (
    <h2>{props.name}</h2>
  );

}
```

You can write:

```
function Student({ name, age, course }) {

  return (

    <div>

      <h2>{name}</h2>

      <p>{age}</p>

      <p>{course}</p>

    </div>

  );

}
```

This is cleaner and commonly used in React applications.

Interview Questions

1. What is a React Component?

A React component is a reusable piece of UI that returns JSX and can be used multiple times in an application.

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2. What are Props?

Props (Properties) are read-only values passed from a parent component to a child component.

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3. Can a child component modify props?

No. Props are **immutable (read-only)**. If data needs to change, it should typically be managed by the parent using state.

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4. Difference between Props and State

Props	State
Passed from parent	Managed inside the component
Read-only	Can be updated
Used for communication	Used for dynamic data

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5. Why are components important in React?

Components make the application modular, reusable, easier to test, and easier to maintain.

Real-World Example

Imagine an e-commerce website:

Instead of creating separate HTML for every product, you create one Product component:

```
<Product name="Laptop" price={60000} />

<Product name="Mobile" price={25000} />

<Product name="Headphones" price={3000} />
```

The same component is reused with different data, just like the Student component in this exercise.